PoolMate

Detailed user guide - how to do everything PoolMate can do: pool libraries, normalize a plate, dilute samples, generate Eppendorf epMotion worklists, and work out flow cell loading. Everything runs locally in your browser.

Which tab do I use?

Tab	Use it to	Produces
Pool	Combine several libraries into one tube (Tube layout), or normalize each into its own well (Plate layout).	DNA.csv + Buffer.csv (+ worksheet, inputs)
Normalize	Dilute every library to one common concentration, plate to plate.	Normalization_DNA.csv + Normalization_Buffer.csv
Dilute	Dilute one sample or a list - on the bench (tubes) or plate to plate.	dilution list CSV, and dilution_DNA.csv + dilution_Buffer.csv
Load	Work out how much of your pool(s) to load onto an ONT or Illumina flow cell.	loading sheet CSV + bench steps

Each tab is self-contained: its own inputs, its own live graphic, and its own export. The graphic always shows **where each sample goes and how much**.

Pool - combine libraries

1. For each sample enter the **source well**, a **name**, and either **concentration reads (ng/µL) + size (bp)**, or a **direct nM** value. Replicate reads can be separated by commas, semicolons or spaces and are averaged.
2. Set the **Pool calculation mode** (target nM, or total fmol) and the **Total final volume**.
3. Choose the **Layout**: **Tube** pools every sample into one tube; **Plate** puts each sample in its own well. In Plate layout an optional **Out well** per sample chooses where it lands on the output plate.
4. Optionally open **Optional extra pooling parameters** for racks, buffer position, pool well, minimum transfer, rounding, dead volume and tip policy.
5. Click **Generate worklists** and download the files.

The three Pool actions

Button	What it does	Robot files?
Calculate molarities	Converts concentration + size to nM from the sample table only.	No
Quick pool table	A manual pipetting table. No fixed target style is required - see below.	No
Generate worklists	The full epMotion export: transfer volumes, buffer, tips, CV check.	Yes

Quick pool target styles

Quick pool does not force one way of specifying the target - pick whichever matches your SOP:

Mode	You enter	Result
Equimolar - fmol per sample	fmol for each sample	Equal molecules from every sample.

Mode	You enter	Result
Equimolar - total fmol	total fmol for the pool	Split equally across the samples.
Equimolar - target nM in a final volume	target nM + final volume (Illumina style)	Same maths as Generate worklists, as a manual table.
Equal volume from each sample	one volume per sample	Simple pooling - the same volume from every sample, so each contributes a different number of molecules (the CV shown reflects this and is expected).

Normalize - one concentration across a plate

1. Choose the **unit** (nM or ng/µL), the **target concentration**, and the **final volume per well**.
2. For each library enter its **input well**, a name, and its **concentration**.
3. Optionally set an **Output well** to place the library elsewhere on the output plate (for example A3 to D5). Blank keeps the same well.
4. Check the live two-plate graphic, then click **Download Normalization worklists**.

Libraries too concentrated to reach the target in one step are flagged - dilute those first, then normalize.

Dilute - one sample or a list

1. Enter each sample's **stock concentration**. Target, final volume, minimum transfer, maximum well capacity and rounding default to the shared values at the top and can be **overridden per row**.
2. Use **Tube view** for bench work (a virtual tube per sample, with a Done checklist) or **Plate view** for a plate-to-plate run.
3. In Plate view set **In well** and **Out well** to control exactly where each sample is diluted.
4. **Download dilution list CSV** for a bench record, or **Download epMotion worklists** to run it on the robot.

Load - flow cell loading

1. Pick the **platform** (Oxford Nanopore or Illumina) and a **flow cell / kit preset**. The target load or loading concentration fills in and stays editable.
2. Choose **One pool** or **Combine several pools**, then enter each pool's concentration (nM, or ng/µL together with the fragment size in bp).
3. With several pools, **Share** sets the relative amount of each - leave every share at 1 for equal molar representation, or set 2 to give a pool twice the share.
4. Click **Calculate loading** for the µL to take from each pool plus the bench steps.

Platform	What you get
Oxford Nanopore	The µL of each pool, the Elution Buffer top-up that fills the library slot, and the loading mix (Sequencing Buffer + Library Beads/Solution + library) to load dropwise via the SpotON port.

Platform	What you get
Illumina	The combined pool molarity, then the dilution steps down to the loading concentration in the final volume - split into serial steps automatically when one step would not be pipettable. The denature step is instrument-specific and is NOT calculated; follow your Denature and Dilute Libraries guide.

Preset starting points

Preset	Default and source note
ONT ligation, < 1 kb	100 fmol (from ~200 fmol input); library to 12 µL in a 75 µL mix (37.5 µL SB + 25.5 µL LIB/LIS)
ONT ligation, 1-10 kb	35-50 fmol (50 used), from 100-200 fmol input; same 75 µL mix
ONT native barcoding (NBD114), MinION/GridION	10-20 fmol (20 used) - lower than the plain ligation kit; same 75 µL mix
ONT native barcoding (NBD114), Flongle	5-10 fmol (10 used); loading much above ~10 fmol can reduce output. Mix = 15 µL SB + 10 µL LIB/LIS + 5 µL library = 30 µL; priming 117 µL FCF + 3 µL FCT
ONT long reads > 10 kb	Load by MASS, not molarity (starting input ~1 µg) - no fmol target
ONT rapid kits (SQK-RBK114)	Specified by mass: 200 ng gDNA per barcode. Use the Custom preset
MiSeq Reagent Kit v3	6-20 pM (20 used); needs at least a 4 nM library; 600 µL
MiSeq Reagent Kit v2	6-10 pM (10 used); 600 µL
NextSeq 500/550 High / Mid Output	1.8 pM / 1.5 pM in 1.3 mL
NextSeq 1000/2000 Standard SBS	Start at 650 pM (onboard denature) or 65 pM (manual); adjust in 100 pM / 10 pM steps
NextSeq 1000/2000 XLEAP-SBS	Start at about 75% of the Standard SBS concentration where there is no specific guidance (e.g. 488 pM vs 650 pM). P1/P2: start at the Standard SBS value; P3/P4 and 600-cycle P1/P2: reduce by about 25%. Titration runs strongly recommended
NovaSeq 6000 / NovaSeq X	Not preset - take the values from the current Denature and Dilute guide for your flow cell and kit, and use Custom

These are vendor starting points and labs routinely optimise loading; figures also differ between kit and protocol versions. Every value is editable and a **Custom** option covers any other flow cell or instrument. Always confirm against your current protocol.

Choosing output wells

Anywhere PoolMate writes to a plate - Pool in Plate layout, Normalize, and Dilute in Plate view - an **Out well** column decides where each sample lands. Leave it blank to keep the sample in its own well. The live graphic shows the move and the exported DNA/Buffer worklists follow it exactly.

Deck, racks and tips

Rack 1 = source plate. **Rack 2** = destination (the pool tube in Tube layout). Buffer is drawn from the reservoir at **position "1"** on the buffer rack, which normally matches the source rack. Tools: **TS_50** = 50 µL tips, **TS_300** = 300 µL tips.

Dilution and pooling cannot share one run

A pooling script executes one DNA transfer list plus one buffer transfer list. Pooling must draw from wells that have already been diluted **and mixed**, so dilution and pooling are always separate runs:

1. **Run 1 - dilute:** use Dilute (or Normalize) to export worklists, run them, then mix the wells.
2. **Run 2 - pool:** generate the pooling worklists drawing from those wells and run them.

Serial (multi-step) dilutions need one run per step for the same reason; PoolMate excludes them from a worklist export and flags them instead.

If a transfer would be below the minimum

Very concentrated libraries can require less than 1 µL, which is not reliably pipettable. Either pre-dilute the sample first (Dilute tab), pick a target/volume that keeps every transfer above the minimum, or set **Low-volume handling** to *pre-dilute hot samples* so PoolMate emits a Dilution.csv plan to run before the pool.

Saving work and file outputs

A plain page reload always starts blank. Use **Save run (JSON)** to keep a run and **Open run** to restore it; if a session was left unsaved a recovery banner offers to restore it. Saved runs include all four tabs.

File	What it is
DNA.csv	Sample/stock transfer worklist for the robot.
Buffer.csv	Buffer/diluent transfer worklist. Load it after the DNA file.
Dilution.csv	Pre-dilution transfers, when low-volume handling produced a plan. Run it first.
*_pooling_worksheet.csv	Full calculation record including the post-rounding CV check.
*_inputs.csv / .txt	The run's settings and sample list; the CSV re-imports into PoolMate.
dilution_list.csv	Bench-readable dilution table (DNA + diluent per sample).
flowcell_loading.csv	Per-pool loading volumes plus the loading summary.

About the numbers

Molarity for dsDNA: **nM = conc(ng/µL) x 10^6 / (660 x size_bp)**. Pool transfers are calculated as equal fmol per sample, rounded to the selected increment, then audited for post-rounding deviation (CV%). Dilutions use **C1V1 = C2V2**. For loading, **fmol = nM x µL**, and an Illumina target of X pM in V µL needs **X x V / 1000** fmol.

Important: platform presets and flow cell loading defaults are vendor starting points. Confirm all volumes, loading concentrations and targets against your own kit / instrument SOP before running the robot or loading a flow cell.